Sebastian Lopez Silva

AI Engineer

Paris, France | sebas.lopezsilva147@gmail.com | +33 07 59 32 58 83 | slopezsi | Sebls | Sebls.github.io

Profile

***AI Engineer**, with solid experience in software engineering and the **production deployment of AI services** for text, image-based and **multimodal** applications. Particularly interested in **LLM systems**, **scalable AI platforms**, and production-grade intelligent systems.*

Education

École des Mines de Paris - PSL, **Engineering Degree** (M.Sc.) (GPA: 4.0/4.0) — 2025 - Expected 2027
Universidad Nacional de Colombia, **Computer Engineering** (B.Sc.) (GPA: 4.0/4.0) — 2021 - Expected 2027

Experience

AI Engineer - *Engin AI* (Jan 2026 – May 2026) – Miami, United States (Independent Contractor)

- Designed and implemented an **AI agent** for **deepfake classification**, orchestrating **inference** across private classification models.
- Developed the **agent architecture** and tool ecosystem, integrating **OpenKB**-based long-term memory, short-term conversational memory, and structured model-interaction workflows.
- Designed and executed **production evaluations** to assess classification performance, agent decision-making, and reliability across **multimodal** deepfake detection scenarios.

Machine Learning Engineer - *Engin AI* (Nov 2025 – Jan 2026) – Miami, United States (Independent Contractor)

- Designed and deployed over 5 production-ready **multimodal inference endpoints** (audio, image, video), processing approximately 4,000 daily requests with an average latency of 126 ms.
- Extended **deep learning** architectures for **audio detection**, processing more than 1 million samples and improving experiment reproducibility by 40%.
- Built **scalable data pipelines** for datasets exceeding 100 GB, reducing deployment time by 60% through automated validations and configuration-driven workflows.

Software Engineer - *Engin AI* (Sept 2025 – Nov 2025) – Miami, United States (Independent Contractor)

- Reduced **ML pipeline** latency by 70-80% by redesigning data preprocessing workflows and optimizing resource utilization in containerized environments.
- Automated annotation validation and debugging workflows (**YOLO/COCO**) via Python scripts, eliminating 75% of manual QA steps and improving dataset reliability.

Junior Software Engineer - *Dataconstructors AI* (Mar 2025 – Sept 2025) – Bogotá, Colombia

- Reduced geospatial processing time from one week to three hours by designing automated orchestration of Python pipelines interconnected with database-backed services.
- Designed **AI-assisted pipelines** for structured technical documentation and validation workflows.
- Developed Python scripts integrating **LLMs** for **semantic extraction** and automatic validation of critical fields, improving database consistency and decreasing operational errors by 60%.
- Developed and maintained backend services integrating external APIs and relational databases (MySQL/PostgreSQL), applying clean architecture principles to ensure maintainability and scalability.

Projects

PhraseCrack — *Personal Project* (2026)

- Developed an open-source daily word-guessing game with **semantic similarity** scoring applying **transformer-based embeddings**.
- Implemented a clean, domain-driven full-stack architecture using Next.js (TypeScript) for the frontend and FastAPI (Python) for the backend, including automated tests and clear documentation.

Skills

Programming Languages: Python • TypeScript • C/C++
Frameworks & Libraries: FastAPI • React • Next.js • Pandas • NumPy • PyTorch • TensorFlow
Databases: SQL (PostgreSQL, SQLite) • NoSQL (Redis, MongoDB)
Infrastructure & Systems: Linux • Git • GitHub Actions • Testing • CI/CD • Docker
Cloud: Google Cloud Platform (Compute Engine, Cloud Storage, Vertex AI)
Artificial Intelligence: LLM (OpenAI API, OpenRouter, Claude Platform) • Computer Vision (segmentation, object detection, classification) • Deep Learning (CNNs, embeddings) • Model evaluation, ablation studies & reproducibility
Languages: Spanish (C2) • English (C1) • French (C1) • Portuguese (A2)